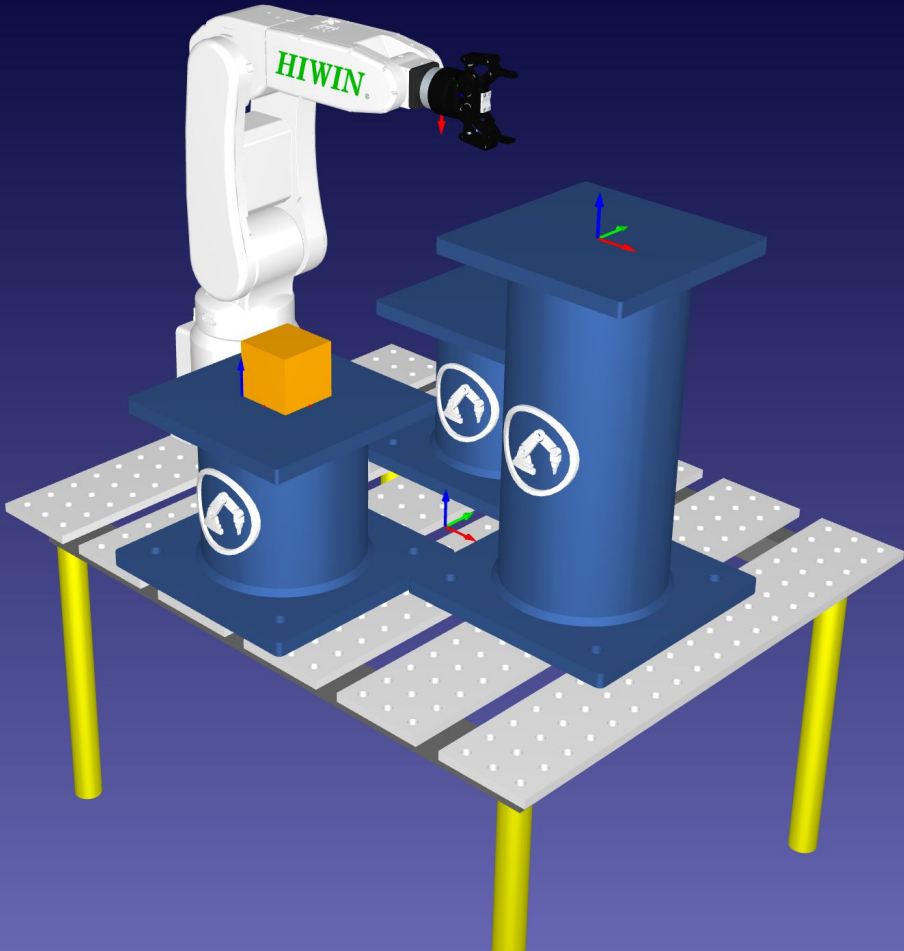




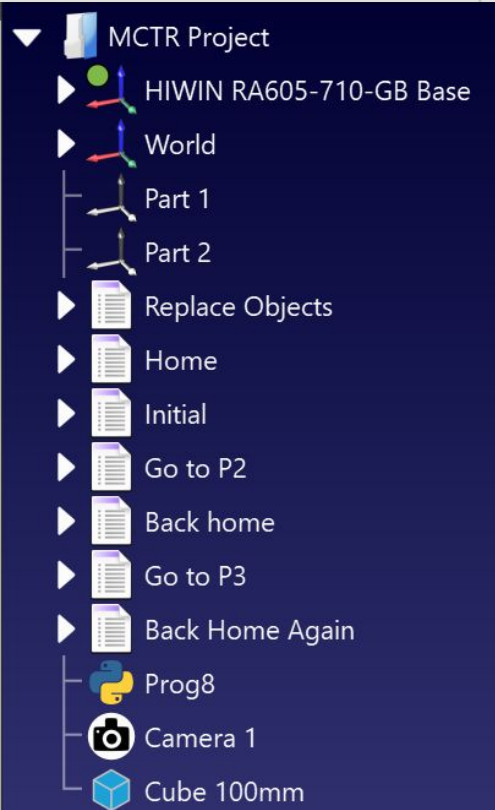
Pick and Place using



RoboDK

Isaac Cejas
and
Totus Tuus Keely

Pick and Place Demonstration



Reference Frames

Robot Base

Tool Frame

RobotiQ 2F-85 Gripper (Open)

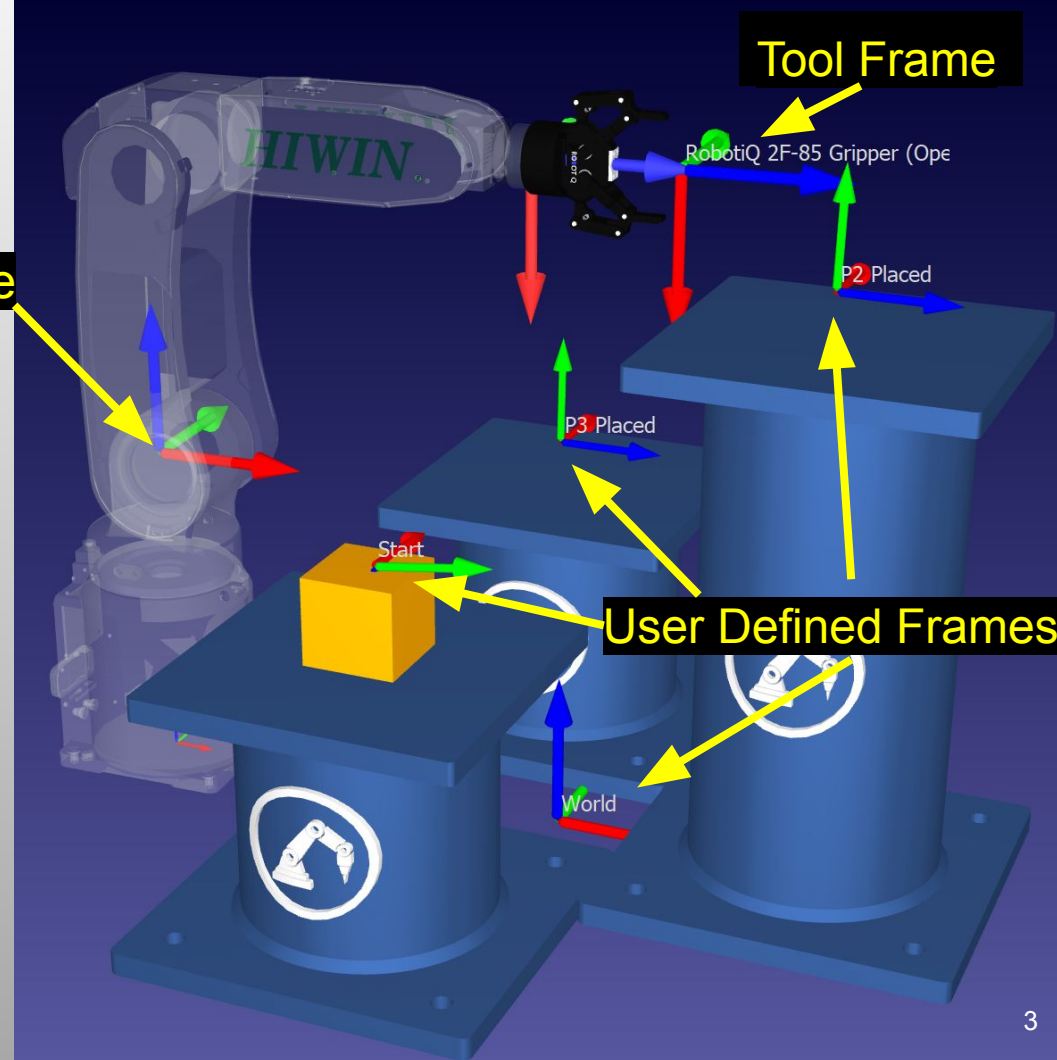
P2 Placed

P3 Placed

Start

User Defined Frames

World

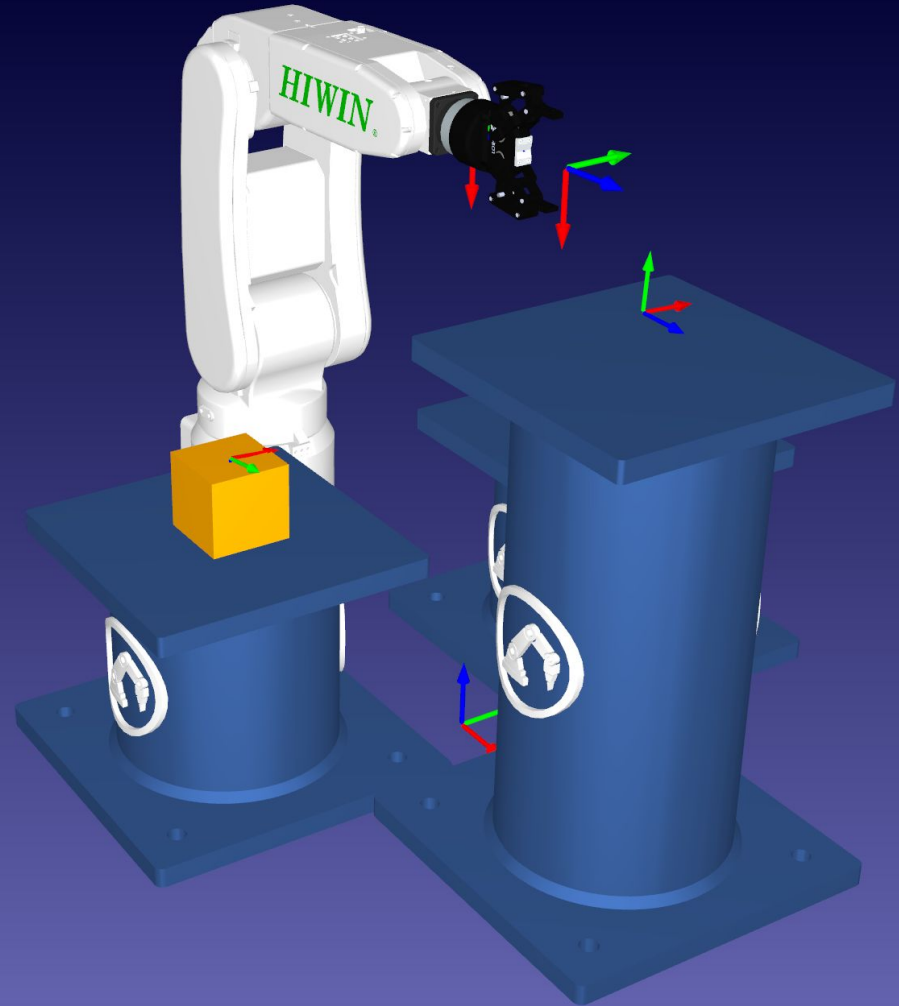


Initial Pose

Initialization

- ▶ Replace Objects
- ▶ Home
- ▶ Initial
- ▶ Go to P2
- ▶ Back home
- ▶ Go to P3
- ▶ Back Home Again

- ▼ Replace Objects
 - Move objects
- ▼ Home
 - Set Ref.: HIWIN RA605-710-GB Base
 - Set Tool: RobotiQ 2F-85 Gripper (Open)
 - MoveJ (Home)

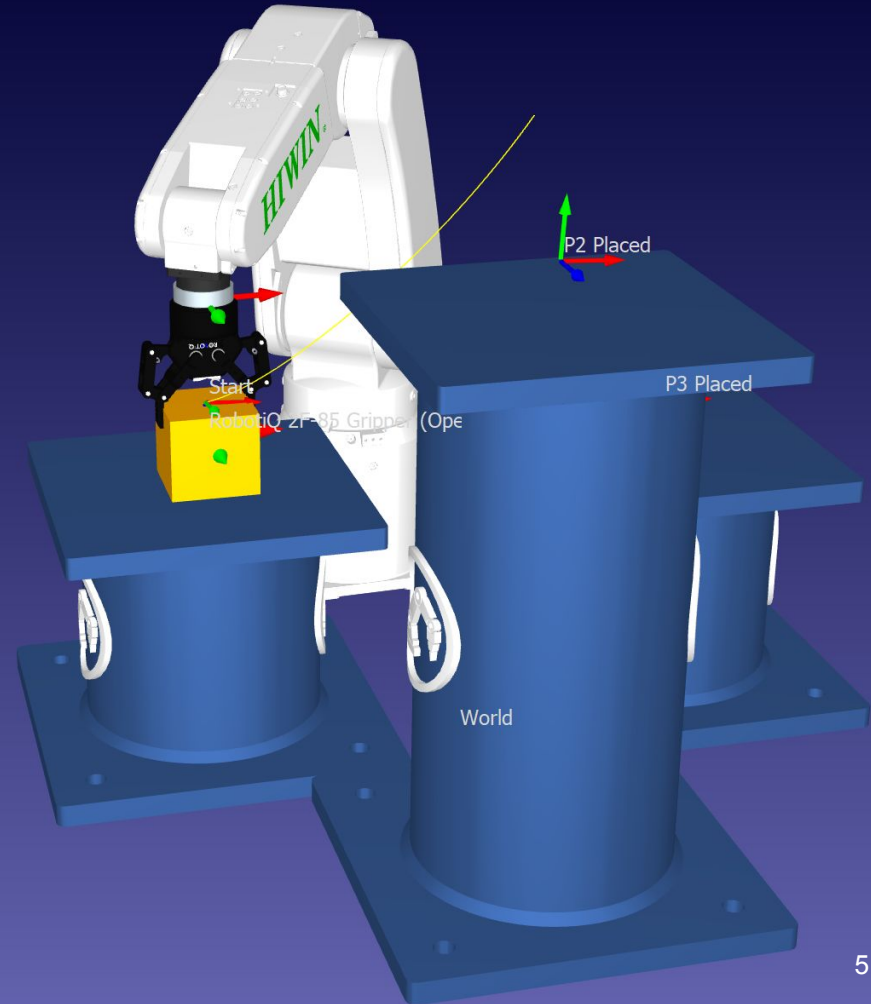


Pedestal 1

- ▶ Replace Objects
- ▶ Home
- ▶ Initial
- ▶ Go to P2
- ▶ Back home
- ▶ Go to P3
- ▶ Back Home Again

▼ Initial

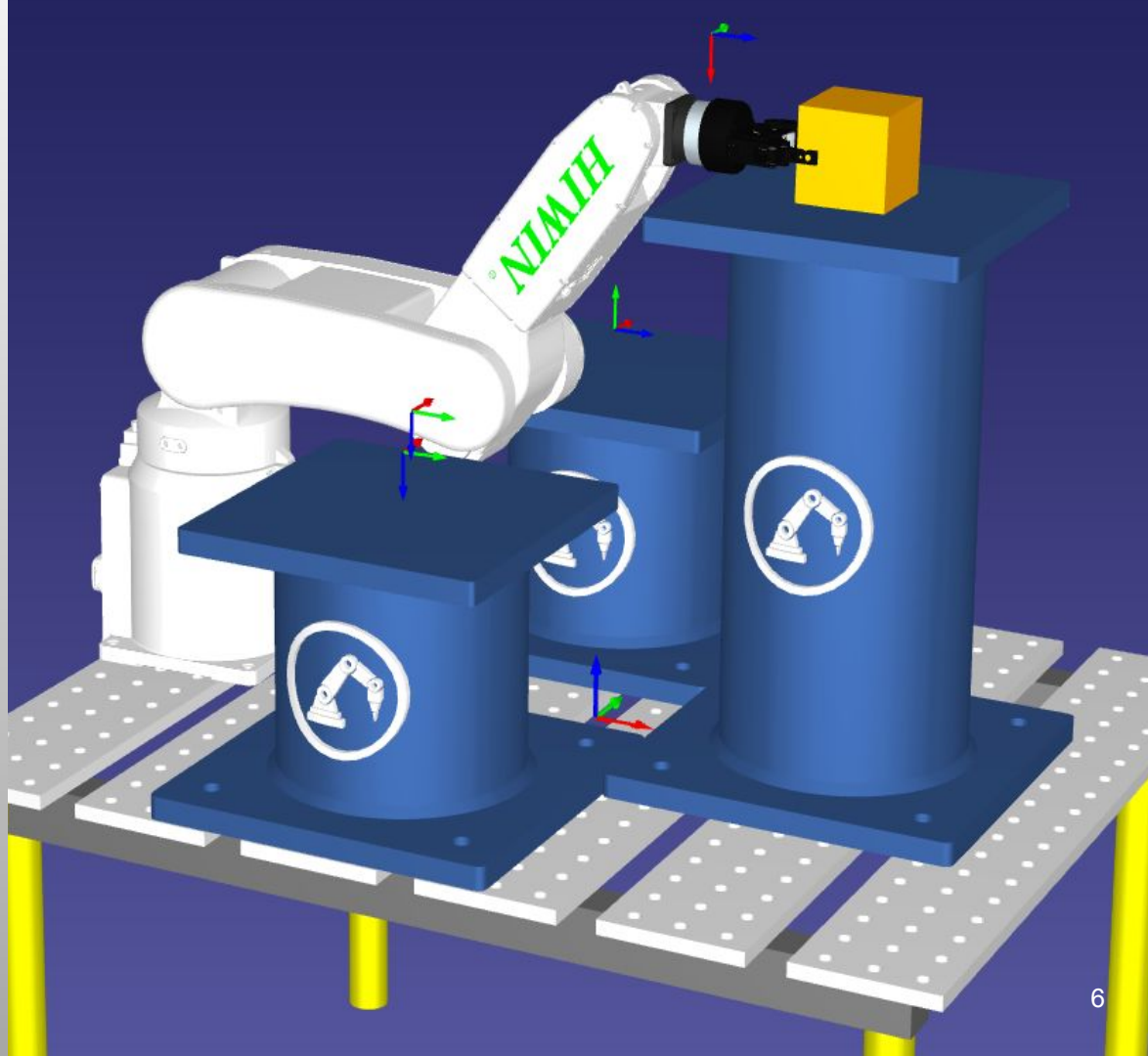
- Set Ref.: HIWIN RA605-710-GB Base
- Set Tool: RobotiQ 2F-85 Gripper (Open)
- MoveJ (Start)
- MoveJ (Target 3)



Intermediate Pose

- ▶ Replace Objects
- ▶ Home
- ▶ Initial
- ▶ Go to P2
- ▶ Back home
- ▶ Go to P3
- ▶ Back Home Again

- Go to P2
 - Attach to RobotiQ 2F-85 Gripper (Open)
 - MoveJ (place block at p1)
- Back home
 - Detach from RobotiQ 2F-85 Gripper (...)
 - Set Ref.: HIWIN RA605-710-GB Base
 - Set Tool: RobotiQ 2F-85 Gripper (Open)
 - MoveJ (Target 10)
 - MoveJ (Home)

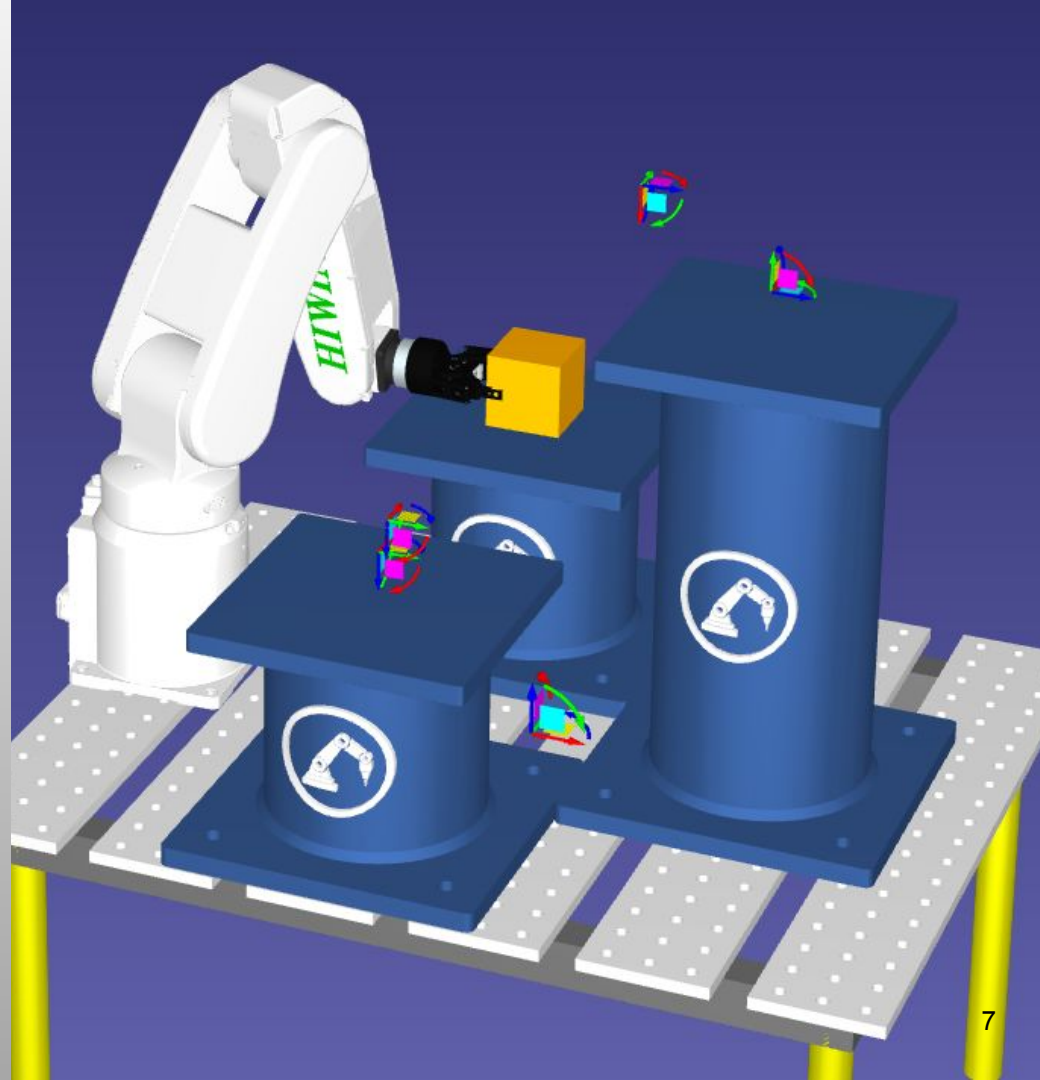


Final Pose

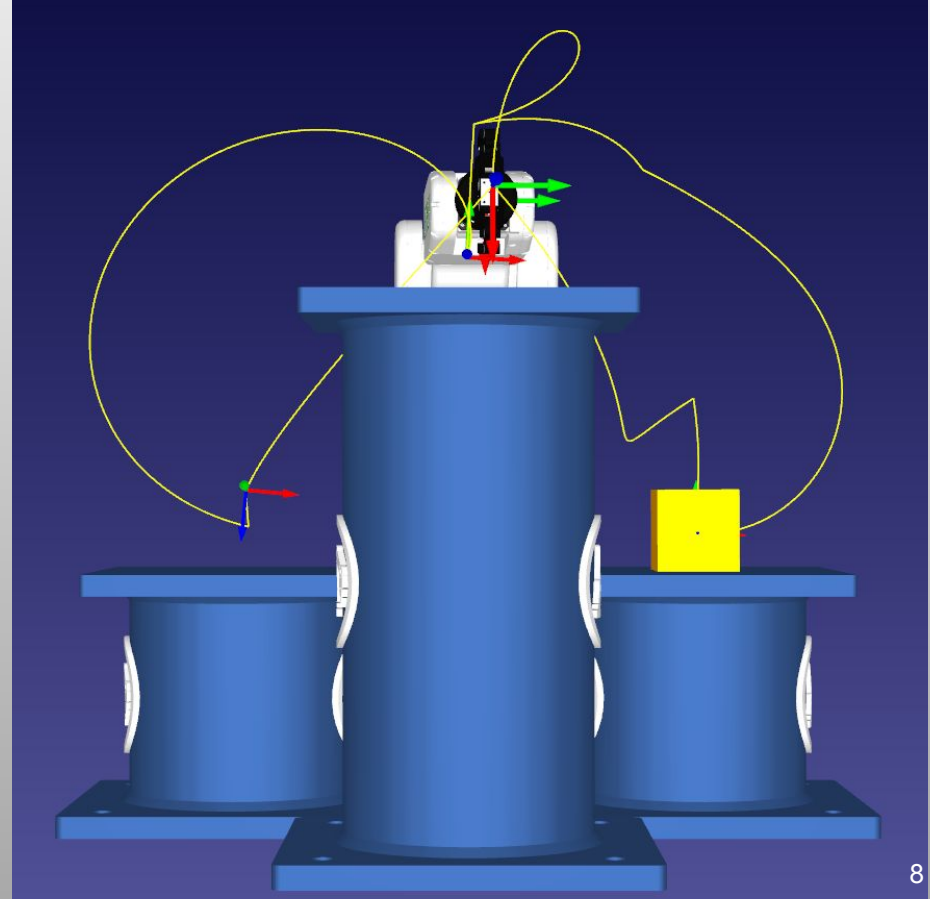
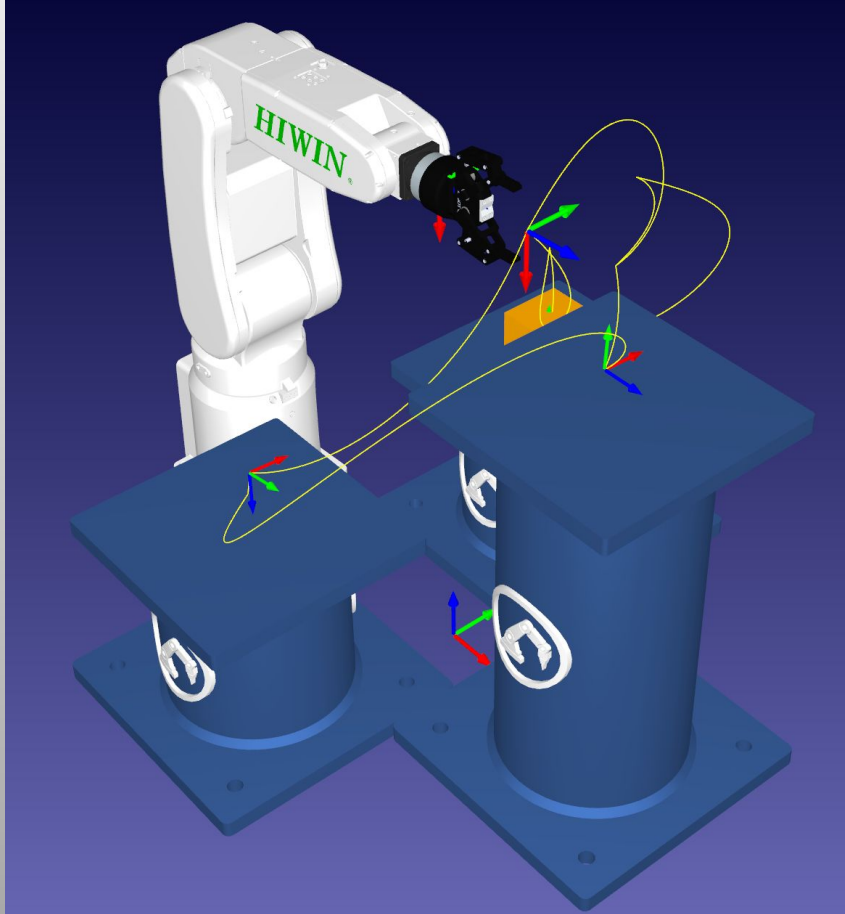
- ▶ Replace Objects
- ▶ Home
- ▶ Initial
- ▶ Go to P2
- ▶ Back home
- ▶ Go to P3
- ▶ Back Home Again

Go to P3

- Set Ref.: HIWIN RA605-710-GB Base
- Set Tool: RobotiQ 2F-85 Gripper (Open)
- MoveJ (Target 10)
- MoveJ (Target 6)
- Attach to RobotiQ 2F-85 Gripper (Open)
- MoveJ (Target 10)
- MoveJ (Target 12)
- MoveJ (Target 7)



End-Effector Path



```

1 from robodk import robolink, robomath as rm
2 import time
3
4 # -----
5 # Initialize RoboDK
6 # -----
7 RDK = robolink.Robolink()
8
9 ROBOT_NAME = 'HIWIN RA605-710-GB'
10 TOOL_NAME = 'RobotiQ 2F-85 Gripper (Open)'
11
12 robot = RDK.Item(ROBOT_NAME, robolink.ITEM_TYPE_ROBOT)
13 if not robot.Valid():
14     raise Exception("Robot not found")
15
16 tool = RDK.Item(TOOL_NAME, robolink.ITEM_TYPE_TOOL)
17 if tool.Valid():
18     robot.setTool(tool)
19
20
21 # -----
22 # Pose acquisition function
23 # -----
24 def get_base_to_tool_matrix(robot):
25     """
26     Returns transformation: Base -> Tool (TCP)
27     """
28     T_base_flange = robot.Pose()
29     T_flange_tool = robot.PoseTool()
30     return T_base_flange * T_flange_tool
31
32
33 # -----
34 # Program execution helper
35 # -----
36 def run_program(rdk, name):
37     prog = rdk.Item(name, robolink.ITEM_TYPE_PROGRAM)
38
39     if not prog.Valid():
40         raise Exception(f"Program '{name}' not found")
41
42     print(f"Running: {name}")
43     prog.RunProgram()
44
45     while prog.Busy():
46         time.sleep(0.1)
47

```

Python Code

```

48 # -----
49 # Run sequence + capture poses
50 # -----
51 pose_log = {}
52
53 program_sequence = [
54     "Replace Objects",
55     "Home",
56     "Initial",
57     "Go to P2",
58     "Back home",
59     "Go to P3",
60     "Back Home Again"
61 ]
62
63 capture_points = {
64     "Home": "Home",
65     "P1": "Initial",
66     "P2": "Go to P2",
67     "P3": "Go to P3"
68 }
69 # Run initial setup
70 run_program(RDK, "Replace Objects")
71
72 # Execute programs and capture poses
73 for label in program_sequence:
74     run_program(RDK, label)
75
76     if label in capture_points.values():
77         pose_log[label] = get_base_to_tool_matrix(robot)
78
79 print("\nAll programs completed.")
80 # -----
81 # Print absolute poses
82 # -----
83 print("\nTransformations (Base -> TCP):")
84 for key, pose in pose_log.items():
85     print(f"\n{key}: \n{pose}")
86
87 # -----
88 # Compute relative transforms
89 # -----
90 def relative_transform(A, B):
91     return rm.invH(A) * B
92
93 print("\nRelative Transforms:")
94
95 if "Home" in pose_log and "Initial" in pose_log:
96     print("\nHome -> P1:\n", relative_transform(pose_log["Home"], pose_log["Initial"]))
97
98 if "Initial" in pose_log and "Go to P2" in pose_log:
99     print("\nP1 -> P2:\n", relative_transform(pose_log["Initial"], pose_log["Go to P2"]))
100
101 if "Go to P2" in pose_log and "Go to P3" in pose_log:
102     print("\nP2 -> P3:\n", relative_transform(pose_log["Go to P2"], pose_log["Go to P3"]))

```

Python Summary

1. Reset
2. Get Matrix
3. Pedestal 1
4. Get Matrix
5. Pedestal 2
6. Get Matrix
7. Wait
8. Pedestal 3
9. Get Matrix
10. Wait
11. Display Matrix
12. Calculate homogeneous Transfer Matrix between points

Location of Points from Robot Base

Transformations (Base -> TCP):

Home:

```
Pose(774.500, 0.000, 380.000, 0.000, 90.000, -0.000):  
[[ -0.000, -0.000, 1.000, 774.500 ],  
 [ -0.000, 1.000, 0.000, 0.000 ],  
 [ -1.000, -0.000, -0.000, 380.000 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

Initial:

```
Pose(384.897, -289.768, -176.088, 179.653, 0.439, -90.649):  
[[ -0.011, 1.000, 0.008, 384.897 ],  
 [ 1.000, 0.011, -0.006, -289.768 ],  
 [ -0.006, 0.008, -1.000, -176.088 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

Output of Points

Go to P2:

```
Pose(947.782, -45.668, 279.573, 102.750, 87.270, -15.770):  
[[ 0.046, 0.013, 0.999, 947.782 ],  
 [ 0.998, 0.052, -0.046, -45.668 ],  
 [-0.053, 0.999, -0.011, 279.573 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

Go to P3:

```
Pose(560.353, 253.821, -22.924, 85.100, 87.372, 1.861):  
[[ 0.046, -0.001, 0.999, 560.353 ],  
 [ 0.998, 0.053, -0.046, 253.821 ],  
 [-0.053, 0.999, 0.004, -22.924 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

Transfer Matrices

```
def relative_transform(A, B):  
    return rm.invH(A) * B
```

Relative Transforms:

Home → P1:

```
Pose(556.088, -289.768, -389.603, 38.330, 89.440, 51.020):  
[[ 0.006, -0.008, 1.000, 556.088 ],  
 [ 1.000, 0.011, -0.006, -289.768 ],  
 [-0.011, 1.000, 0.008, -389.603 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

Transfer Matrices

P1 → P2:

```
Pose(234.906, 569.069, -452.804, -88.941, -3.307, -2.646):  
[[ 0.997, 0.046, -0.058, 234.906 ],  
 [ 0.057, 0.021, 0.998, 569.069 ],  
 [ 0.047, -0.999, 0.018, -452.804 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

P2 → P3:

```
Pose(297.007, -291.383, -397.721, -0.828, -0.000, 0.000):  
[[ 1.000, -0.000, -0.000, 297.007 ],  
 [ 0.000, 1.000, 0.014, -291.383 ],  
 [ 0.000, -0.014, 1.000, -397.721 ],  
 [ 0.000, 0.000, 0.000, 1.000 ]]
```

Hand Proofs

Home to Pedestal 1

$$A = \begin{bmatrix} 0 & 0 & 1 & 774.5 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 380 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} .006 & -.008 & 1 & 556.088 \\ 1 & 0.011 & -.006 & -289.768 \\ -0.011 & 1 & 0.008 & -389.603 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -0.011 & 1 & 0.008 & 384.897 \\ 1 & 0.011 & -0.006 & -289.768 \\ -0.006 & 0.008 & -1 & -176.088 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \right)$$

Hand Proofs

Pedestal 1 to Pedestal 2

$$A = \begin{bmatrix} -.011 & 1 & 0.008 & 384.897 \\ 1 & 0.011 & -.006 & -289.768 \\ -.006 & 0.008 & -1 & -176.088 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} .997 & 0.046 & -.058 & 234.906 \\ 0.057 & 0.021 & .998 & 569.069 \\ 0.047 & -.999 & 0.018 & -452.804 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0.046409 & 0.012502 & 0.998782 & 947.76 \\ 0.997345 & 0.052225 & -0.04713 & -45.8854 \\ -0.052526 & 0.998892 & -0.009668 & 279.859 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \right)$$

Hand Proofs

Pedestal 2 to Pedestal 3

$$A = \begin{bmatrix} .046 & .013 & .999 & 947.782 \\ .998 & .052 & -.046 & -45.668 \\ -.053 & .999 & -.011 & 279.573 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 & 297.007 \\ 0 & 1 & .014 & -291.383 \\ 0 & -.014 & 1 & -397.721 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0.046 & -9.86 \times 10^{-4} & 0.999182 & 560.333 \\ 0.998 & 0.052644 & -0.045272 & 253.888 \\ -0.053 & 0.999154 & 0.002986 & -22.8851 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{Ⓢ}$$

Orthogonality of P1 -> P2 Rotation Matrix

$$A = \begin{bmatrix} 0.997 & 0.046 & -0.058 \\ 0.057 & 0.021 & 0.998 \\ 0.047 & -0.999 & 0.018 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 0.999467 & 1.06 \times 10^{-4} & -9.4 \times 10^{-5} \\ 1.06 \times 10^{-4} & 1.00056 & 3.08 \times 10^{-4} \\ -9.4 \times 10^{-5} & 3.08 \times 10^{-4} & 0.999692 \end{bmatrix}$$

$\approx$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Determinant condition

$$A = \begin{bmatrix} 0.997 & 0.046 & -0.058 \\ 0.057 & 0.021 & 0.998 \\ 0.047 & -0.999 & 0.018 \end{bmatrix}$$

$$\det(A) = 0.99985828 \approx 1$$

Any
Questions?